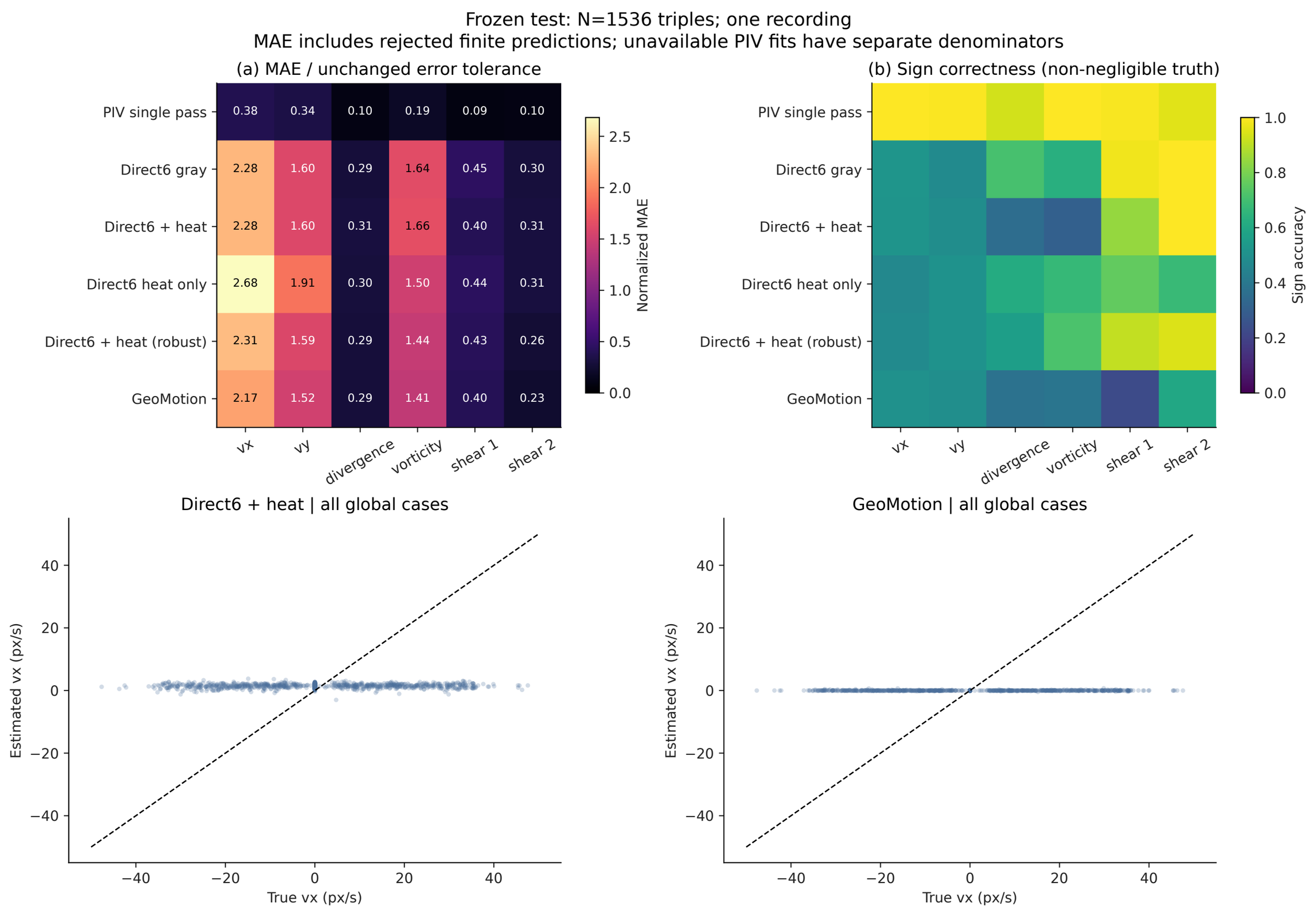

Frozen test: N=1536 triples; one recording
MAE includes rejected finite predictions; unavailable PIV fits have separate denominators
(a) MAE / unchanged error tolerance
PIV single pass
Direct6 gray
Direct6 + heat
Direct6 heat only
Direct6 + heat (robust)
GeoMotion
0.38 0.34 0.10 0.19 0.09 0.10
2.28 1.60 0.29 1.64 0.45 0.30
2.28 1.60 0.31 1.66 0.40 0.31
2.68 1.91 0.30 1.50 0.44 0.31
2.31 1.59 0.29 1.44 0.43 0.26
2.17 1.52 0.29 1.41 0.40 0.23
vx
vy
divergence
vorticity
shear 1
shear 2
Normalized MAE
0.0
0.5
1.0
1.5
2.0
2.5
(b) Sign correctness (non-negligible truth)
Sign accuracy
0.0
0.2
0.4
0.6
0.8
1.0
Direct6 + heat | all global cases
GeoMotion | all global cases
Estimated vx (px/s)
True vx (px/s)
−40
−20
0
20
40